

Revision Questions

Revision Question 1

Graphical Method- Music Festival Ticket Planning Problem

A company is organizing a large outdoor music festival and sells two types of tickets:

- VIP tickets
- Regular tickets

Each ticket requires staffing and security resources as shown below:

Ticket Type	Staffing Hours	Security Units	Profit (£)
VIP	3	2	50
Regular	2	4	40

The company has:

- 60 staffing hours available
- 80 security units available

The objective is to determine how many VIP and Regular tickets should be sold in order to **maximize total profit**.

Tasks

1. Clearly define the decision variables.
2. Formulate the objective function.
3. Formulate all constraints.
4. Solve the problem using the graphical method.
5. Identify all corner (extreme) points of the feasible region.

6. Determine the optimal solution and the maximum profit.
7. Identify which constraints are binding at the optimal solution.

Note: Clearly show your working, including intercept calculations and objective function evaluation at corner points.

Revision Question 2

Sensitivity Analysis – Dessert Production Problem

A dessert manufacturing company produces two premium desserts:

- Strawberry Delight
- Mango Supreme

Each dessert requires fruit concentrate and sugar as key ingredients.

Product	Fruit Units	Sugar Units	Profit (£)
Strawberry (x_1)	2	1	8
Mango (x_2)	1	3	6

Available resources:

- 40 units of fruit concentrate
- 45 units of sugar

The company wants to determine the optimal production mix that maximizes profit.

Tasks

1. Formulate the Linear Programming model clearly defining:
 - Decision variables
 - Objective function
 - Constraints
2. Solve the model using Excel Solver.
3. Produce a Sensitivity Report and answer the following:
 - (a) Identify which constraints are binding.
 - (b) What are the shadow prices of the resources?
 - (c) If the profit of Mango increases from £6 to £7, will the optimal solution change? Explain using allowable range.
 - (d) If 5 additional units of fruit become available, what happens to total profit?
 - (e) If the company can purchase 10 additional units of sugar for £15, should it do so? Justify your answer.
4. Interpret the managerial meaning of the shadow price for fruit concentrate.

Instruction: Clearly interpret all sensitivity results in business terms.

Revision Question 3

Transportation Problem – Laptop Distribution Network

A technology company operates two manufacturing plants that supply laptops to three regional distribution warehouses.

The monthly supply and demand quantities are as follows:

- Plant 1 supply: 70 units
- Plant 2 supply: 50 units
- Warehouse 1 demand: 40 units
- Warehouse 2 demand: 30 units
- Warehouse 3 demand: 50 units

The transportation cost per unit (in £) is given in the table below:

	W1	W2	W3
Plant 1	4	6	8
Plant 2	5	3	7

Tasks

1. Define appropriate decision variables.
2. Formulate the Linear Programming model clearly stating:
 - Objective function
 - Supply constraints
 - Demand constraints
 - Non-negativity constraints
3. Verify whether the transportation problem is balanced.
4. Solve the problem using Excel Solver.
5. Determine:
 - The optimal shipping plan
 - The minimum total transportation cost
 - Which routes (if any) are unused

6. Briefly explain why transportation problems often produce integer solutions even without integer constraints.

Instruction: Clearly show the structure of the supply–demand network and interpret the managerial meaning of your solution.

Revision Question 4

Minimum Cost Flow Problem – Distribution Through a Transshipment Hub

A company distributes products from two supply centres to two customer zones. However, all shipments must pass through a central transshipment hub.

Supply Nodes:

- S_1 : 30 units
- S_2 : 20 units

Demand Nodes:

- D_1 : 25 units
- D_2 : 25 units

Transshipment Node:

- T : Flow in must equal flow out

Shipping Costs per Unit (£):

Arc	Cost
$S_1 \rightarrow T$	2
$S_2 \rightarrow T$	3
$T \rightarrow D_1$	4
$T \rightarrow D_2$	5

Tasks

1. Define decision variables for each arc in the network.
2. Formulate the minimum cost flow model including:
 - Objective function
 - Supply constraints
 - Demand constraints
 - Flow balance constraint at node T
 - Non-negativity constraints

3. Solve the model using Excel Solver.
4. Determine:
 - The optimal flow on each arc
 - The minimum total transportation cost
 - Which arc carries the largest amount of flow
5. Explain the role of flow balance constraints in network optimisation models.

Instruction: Clearly represent the network structure and interpret the economic meaning of the optimal solution.

Revision Question 5

Maximum Flow Problem – Water Distribution Network

A water authority must determine the maximum amount of water that can be delivered from a source reservoir to a city.

The network structure is:

- Source $\rightarrow A$ (capacity 20)
- Source $\rightarrow B$ (capacity 30)
- $A \rightarrow$ Sink (capacity 25)
- $B \rightarrow$ Sink (capacity 15)
- $A \rightarrow B$ (capacity 10)

Tasks

1. Define the decision variables.
2. Formulate the maximum flow model including:
 - Objective function
 - Capacity constraints
 - Flow balance constraints
3. Solve using Excel Solver.
4. Determine the maximum possible flow.
5. Identify the bottleneck arc.
6. Explain how increasing a capacity might affect the solution.

Revision Question 6

Assignment Problem – Developer Allocation

A software company must assign 4 developers to 4 projects. Each developer must be assigned to exactly one project.

The cost matrix (in £ thousands) is:

	P1	P2	P3	P4
D1	9	2	7	8
D2	6	4	3	7
D3	5	8	1	8
D4	7	6	9	4

Tasks

1. Define binary decision variables.
2. Formulate the assignment model.
3. Solve using Excel Solver.
4. Determine the minimum total cost.
5. Explain why assignment problems often yield integer solutions without explicitly imposing integrality.

Revision Question 7

Knapsack Problem – Startup Project Selection

A startup has a budget of £15 million and must choose projects to maximize profit.

Project	Cost	Profit
A	6	10
B	5	8
C	4	6
D	7	11

Tasks

1. Define binary decision variables.
2. Formulate the knapsack model.
3. Solve using Excel Solver.
4. Compare the solution with the LP relaxation.
5. Explain whether rounding the LP solution gives the optimal integer solution.

Revision Question 8

Multiple Knapsack Problem – Truck Loading

A logistics company has two trucks:

- Truck 1 capacity = 10
- Truck 2 capacity = 8

Items available:

Item	Weight	Profit
1	4	10
2	6	12
3	5	9
4	3	6

Tasks

1. Define decision variables.
2. Formulate the multiple knapsack model.
3. Solve using Excel Solver.
4. Determine maximum profit.
5. Explain how this differs from the single knapsack problem.

Revision Question 9

Generalised Assignment Problem – Workforce Scheduling

A company has 3 workers and 5 tasks. Each task must be assigned to exactly one worker.

Worker capacity (hours available):

- Worker 1: 8 hours
- Worker 2: 10 hours
- Worker 3: 7 hours

Each task requires different hours depending on the worker assigned.

Tasks

1. Define decision variables.
2. Formulate the generalised assignment model.
3. Include:
 - Assignment constraints
 - Capacity constraints
 - Objective function
4. Solve using Excel Solver.
5. Explain how this differs from the classical assignment problem.

Revision Question 10

Bin Packing Problem – Warehouse Storage

A warehouse must pack items into bins of capacity 10.

Items to pack:

2, 5, 4, 7, 1, 3, 8

Tasks

1. Define binary decision variables.
2. Formulate the bin packing model.
3. Minimize the number of bins used.
4. Solve using Excel Solver.
5. Compare your optimal solution with the First-Fit heuristic solution.

Revision Question 11

Goal Programming – Sustainable Production Planning

A factory produces two products:

$$x_1 = \text{Product A}$$

$$x_2 = \text{Product B}$$

Profit:

$$Z = 8x_1 + 6x_2$$

Labor constraint:

$$4x_1 + 2x_2 \leq 100$$

Management has the following goals:

- Achieve a profit target of 200.
- Produce at least 30 units of Product A.
- Avoid exceeding 100 labor hours.

Tasks

1. Introduce appropriate deviational variables.
2. Formulate the goal programming model.
3. Clearly state the objective function.
4. Solve using Excel Solver.
5. Interpret the meaning of the deviations.